

1. Factors of a Number

```
n = int(input("Enter a number: "))
```

```
print("Factors are:")
for i in range(1, n + 1):
    if n % i == 0:
        print(i, end=" ")
```

2. Factorial

```
n = int(input("Enter a number: "))
```

```
fact = 1
for i in range(1, n + 1):
    fact *= i
```

```
print("Factorial =", fact)
```

3. Palindrome Number

```
n = input("Enter a number: ")
```

```
if n == n[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```

4. Perfect Number

```
n = int(input("Enter a number: "))
```

```
s = 0
for i in range(1, n):
    if n % i == 0:
        s += i
```

```
if s == n:
    print("Perfect Number")
else:
    print("Not Perfect Number")
```

6. Fibonacci Series

```
n = int(input("Enter number of terms: "))
```

```
a, b = 0, 1
```

```
for i in range(n):  
    print(a, end=" ")  
    a, b = b, a + b
```

7. Happy Number

```
n = int(input("Enter a number: "))
```

```
while n != 1 and n != 4:
```

```
    s = 0
```

```
    while n > 0:
```

```
        d = n % 10
```

```
        s += d * d
```

```
        n //= 10
```

```
    n = s
```

```
if n == 1:
```

```
    print("Happy Number")
```

```
else:
```

```
    print("Not Happy Number")
```

8. Prime Number

```
n = int(input("Enter a number: "))
```

```
count = 0
```

```
for i in range(1, n + 1):
```

```
    if n % i == 0:
```

```
        count += 1
```

```
if count == 2:
```

```
    print("Prime Number")
```

```
else:
```

```
    print("Not Prime Number")
```

9. Armstrong Number

```
n = int(input("Enter a number: "))
```

```
temp = n
```

```
s = 0
```

```
while temp > 0:
```

```
    d = temp % 10
```

```
    s += d ** 3
```

```
    temp //= 10
```

```
if s == n:
```

```
    print("Armstrong Number")
```

```
else:
```

```
    print("Not Armstrong Number")
```